Proceedings of the 41st Chinese Control Conference
July 25-27, 2022, Hefei, China

LQR-based Balance Control of Two-wheeled Legged Robot

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Abstract: Recently, the research of wheel-legged robot (WLR) has become a hot spot. WLR has fast speed, high efficiency and strong adaptability to terrain, which makes full use of the advantages of wheeled robots and leg robots. This paper presents a successful case of balance control applied to an actual WLR platform. The mechanical structure is firstly introduced in this paper. Subsequently, the simplified dynamic model of the robot is described. Based on the linear quadratic regulator (LQR), the balance controller is constructed according to the established dynamic equation. Both simulation and experiments are carried out, which sufficiently demonstrate the efficiency of the designed LQR controller.

Key Words: LQR balance control, underactuated robotics optimizion, two-wheeled legged robot

1 Introduction

Nowadays, wheeled robots are becoming more and more popular and occupy a major position in the robot industry [1]. Wheeled robots are highly efficient in execution, but less adaptable to the terrain. Researchers have been working hard to find a wheeled robot that is suitable for a wide range of tasks. However, wheeled robots are hardly comparable to legged robots when it comes to adapting to complex environments. Legged robots have a long history of research, especially for biped robots. At present, this kind of robot has developed to the level of humanoid robot [2], which can adapt to the human working environment and effectively solve many problems. The legged robots are more adaptable to the terrain, but not sufficiently efficient in motion. Dynamic stability is the reason why biped robots require complex control algorithms to balance them. 'ASIMO' from Honda [3], 'QRIO' from Sony [4] , Artrobot from Iron Man [5] are representative examples of biped robots that can walk, run, stand and even dance. These robots maintain stability by planning their steps. A prominent advantage of biped robots is that they have fewer legs, which can ensure lighter weight, less manufacturing cost and more flexibility.

The advantages of wheeled robots and biped robots are complementary to each other. Therefore, WLR has come into being. It not only has the strong terrain adaptability of legged robot, but also has the fast response and execution accuracy of wheeled robots. If the motion balance problem is solved well and artificial intelligence technology is used as the brain, WLR will have broad application scenarios and great commercial value [3]. Specifically, they can not only be widely used in airports, hotels, pension and other service industries, but also have important values in teaching aids, entertainment films, military equipment and other aspects in colleges and universities. To sum up, WLR has high research value because of its strong adaptability and maneuverability.

This work is supported by the National Natural Science Foundation of China under Grant 61873132 and the Opening Project of Guangdong Provincial Key Lab of Robotics and Intelligent System.

Biao Lu is the corresponding author of this paper.

At present, there are many institutions studying WLR. Ascento of Eidgenössische Technische Hoheschule Zürich is now one of the most small WLR [6] . Handle from Boston Dynamic is regarded as one of the leaders in WLR [7] . The research team from South University of Science and Technology in China has successfully built the actual equipment Nezha [8] [9]. The WLR - II designed by Harbin Institute of Technology is a typical example of experiments on the real equipment [10] [11] . Apart from these, Ollie, developed by Tencent, is a new type of WLR [12] .

In order to design an efficient balance control algorithm without losing robustness in the field of WLR, a new type of WLR platform is studied in this paper. The mechanical construction of the WLR platform is first prepared for subsequent balance control requirements. With further research, the platform can also perform more complex tasks. It is forseeable that the designed WLR platform will also be applied to a variety of scenarios, not just limited to the moderate ideal environment. This WLR has now completed the assembly of entities. Compared with some existing results, this platform is more simple and dexterous, which makes it easier to operate in actual experiments.

After the assembly of the actual experimental conditions, the first step is to realize the balance control of WLR. As a reliable scheme, Linear Quadratic Regulator strategy (LQR) is adopted in this paper to realize balance control by using the torque of hub motor as control input [13] - [15] . The balance control scheme is first verified by numerical simulation, and the control parameters are adjusted sufficiently to achieve the desired performance. Besides, The effectiveness of the proposed control strategy is also demonstrated by experiments on the designed actual equipment.

The contributions of this paper are summarized as follows. In Section 2, the electrical components and mechanical structure design of WLR experimental platform are introduced. The dynamic model of the WLR platform is established and linearized in Section 3. The LQR balance controller is then constructed according to the model and established in Section 4. Next, numerical simulation is carried out in matlab and simulink environment in Section 5. Finally, experiments are carried out on an actual equipment and relevant results are analyzed in Section 6.

2 System Introduction

2.1 Mechanical Structure

As shown in Figure 1, this WLR platform combines the advantages of wheeled robots and leg robots. The four-link mechanism is adopted in the mechanical structure, which can realize a series of motion modes and meet the design requirements of different tasks. The upper part of the robot platform is equipped with a micro computer, actuators and other electrical equipments. The lower part is composed of two legs, which are connected with the wheel hub motors. In addition, through the rotation of the joint motor, the height of the robot can be adjusted from 35 cm to 74 cm. This configuration is based on the topology optimization strategy. The portability and strength of WLR constructed with this mechanical structure have been improved in [7] .

3D printing technology is chosen to produce the overall framework. Leg joint motors and wheel hub motors need to be installed on the base support of the robot. The electrical equipment applied to the robot is given in Table 1, including the selected motor model and specifications, onboard computer model, motor driver type and so on. By the way, IMU means inertial measurement unit.

Table 1: Robotic Electrical Equipment

Onboard Computer	NVIDIA JETSON NX NANO
Hub Motor Driver	ZLAC8015D
Robot Battery	24 V
Joint Motor	unitree A1
Connection Spring	1.5 N/m
Hub Motor	ZLLG40ASM100
IMU	HWT-906
Wheel Motor USB to CAN	CANalyst-II

2.2 Hardware and Software

At the hardware level, joint motors and hub motors are two relatively important parts. Regarding the joint motor, the rated torque can reach 17 Nm, and the maximum instantaneous peak value can reach 35 Nm. Encoders are embedded within the wheel hub motors, which help to detect the angular displacement of the wheel. It allows the robot to obtain enough information to improve the stability of the balance control. The leg joints of the robot are equipped with springs, which help to reduce the burden on the joint motors. In terms of communication mode, the joint motor uses the RS-485 serial communication and the hub motor uses CAN communication. The robot is also equipped with an inertial measurement unit (IMU) to detect the initial attitude of the robot. IMU acquires the information to construct the controller. The platform uses 24 V lithium battery for power supply. After practical measurements, the weight of the system is 10.25 kg, and the battery life is about 3.5 h.

A Kalman filter is elaborately designed to fuse the feedback values from the motor and the information obtained by the IMU, so as to provide an estimate of the system state. The WLR is also equipped with a remote controller to support remote operation. The remote controller communicates with the robot via pulse width modulated waves. It can manipulate the retraction of the leg joints as well as the rotation of the wheels. According to the analysis in Subsection 2.1, the jump and balance of the robot can be decoupled. Therefore, the two rockers of the remote controller can control the legs and wheels separately for better performance.

Fig. 1: WLR prototype designed in this paper

3 Modeling

Before proceeding with the controller construction, dynamic model of the WLR should be first derived and analyzed.

3.1 Representative Symbols and Convention Notations

1) In order to facilitate the description of the motion and the establishment of the dynamic model, the following variables are defined : θ_p represents the inclination of the robot with respect to the vertical direction. ω and v indicate the angular velocity and line speed of the wheel hub motor. β shows the inclination angle of the slope for robot movement. L_0 stands for the initial distance between the machine's overall centroid and the axial center of the wheel hub motor.
Other definitions are given in the following Table 2.

Table 2: Other Defined Robot-related Variables

m	quality of the upper part of the robot
m_p	quality of wheel hub motor
τ_k	the torque provided by joint motor
τ_w	the torque provided by wheel hub motor
R	radius of wheel hub motor
D	distance between two wheel hub motors
x	the position where the robot runs
J	the rotational inertia of the wheel hub motor

2) $\boldsymbol{q} = [x\ \theta_p]^T$ is the state vector of the system, which is composed of the robot displacement x and the angle θ_p. Figure 2 is the simplified robot model drawn from the

side, and some symbols of variables are indicated in the figure.

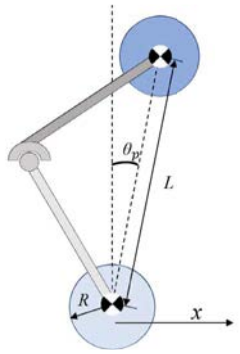

Fig. 2: The simplified WLR model from the side

3.2 Assumptions

In order to facilitate analysis, some reasonable assumptions are adopted:

1) The friction between the connecting rod and the hub motor is ignored.
2) The system delay is small enough to be neglected.

In subsequent experiments, it can be seen that the control strategy is robust and against the above-mentioned disturbances.

3.3 Rigid Body Dynamics Model

In this paper, balance control is carried out under the circumstance of fixed leg lengths. The WLR consists of two wheels on the bottom and an inverted pendulum on top, as shown in Figure 3.

According to the symbols and coordinates specified in Section 3.1, $\boldsymbol{q}$ is taken as the generalized state variables of the system. The following Euler-Lagrange modeling method is used to build the dynamic model:

$$L = T - V \tag{1}$$

All generalized coordinates are independent, the Lagrange equation then can be obtained as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{\boldsymbol{q}}}\right) - \frac{\partial L}{\partial \boldsymbol{q}} = \boldsymbol{J}^T * \boldsymbol{f} \tag{2}$$

In equation (2) , t is a continuous time variable, $\boldsymbol{J}$ is the Jacobian matrix of the whole robot system, and $\boldsymbol{f}$ represents the external force of the system in Cartesian coordinate system. In order to ensure the accuracy of the calculation, the horizontal and vertical force components in $\boldsymbol{f}$ are designed to be zero, and the torque component is designed to be the sum of the torque provided by the motor and spring. With the common form of robot dynamic formulas, the following equation can be derived:

$$M\ddot{\boldsymbol{q}} + C\dot{\boldsymbol{q}} + G = u \tag{3}$$

M, C, and G are auxiliary matrices calculated from the kinetic equation. M is the generalized inertia matrix, G is the gravity matrix, and C repersents the Coriolis force matrix. The explicit expressions of M, C, G, U are given as follows:

$$M = \begin{bmatrix} m + 2m_p + \frac{2J}{R^2} & mLcos(\beta+\theta) \\ mLcos(\beta+\theta) & mL^2 \end{bmatrix} \tag{4}$$

$$C = \begin{bmatrix} 0 & -msin(\beta+\theta)L\dot{\theta} \\ 0 & 0 \end{bmatrix} \tag{5}$$

$$G = \begin{bmatrix} (m+2m_p)gsin\beta \\ -mgLsin\theta \end{bmatrix} \tag{6}$$

$$u = \begin{bmatrix} \frac{\tau_w}{R} \\ -\tau_w \end{bmatrix} \tag{7}$$

The values of the relevant variables in the matrix are measured and calculated according to the actual equipment, which ensures that the simulation results are referable. Figure 3 provides the simplified model of the whole robot system, where the reference direction of the world coordinate system is also depicted. Other parameters are defined in Subsection 3.1.

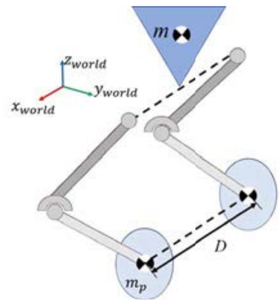

Fig. 3: Linearized WLR model in world coordinate system

4 Control Strategy Analysis

According to the relevant analysis, WLR is essentially an unstable system. With elaborate controller design, the WLR platform is able to perform the specific task. For the balance control problem in this paper, LQR control strategy is adopted.

4.1 System Analysis and Controller Selection

As analyzed in Section 3, after appropriate simplifications, the robot model can be seen as an inverted pendulum with two wheels at the bottom. The LQR control strategy is an optimal control method for achieving control goals at minimal cost. It is an effective scheme for the balance control of WLR. On the premise of ensuring the robustness of the system, the recovering time from disturbances should also be as short as possible. This provides a solid foundation for robots to perform more complex actions.

4.2 Controller Design and Stability Analysis

To proceed with the LQR controller design, the following cost function J is first designed to achieve the goal of optimal control:

$$J = \int_0^\infty (\boldsymbol{q}^T Q \boldsymbol{q} + \boldsymbol{u}^T R \boldsymbol{u})\, dt \tag{8}$$

$\boldsymbol{u}$ is the control vector, which is composed of torques of the left and right hub motors. By designing matrices Q and R, the balance control task of WLR can be accomplished. Besides, the key of LQR is to solve the Riccati equation in discrete time. The following equation presents a part of the derivation process.

$$A^T + PA + Q - PBR^{-1}B^TP = 0 \tag{9}$$

Matrix A and matrix B are the state matrices of the system, and matrix Q and matrix R are given by designer, so the equation contains only matrix P is an unknown quantity. The values of the matrix Q and the matrix R in simulation are given below.

$$Q = diag[100\ 100\ 10\ 10\ 800\ 800] \tag{10}$$

$$R = diag[0.2\ 0.15] \tag{11}$$

In the Riccati equation, the optimal control feedback gain matrix is selected as follows:

$$K = R^{-1}B^TP \tag{12}$$

In this way, the LQR control of the system is completed. Matrix Q and matrix R represent the weights of control and state variables. Usually, matrix Q and matrix R need to be designed as a diagonal matrix. For these weights, repeated tests should be carried out in the simulation environment and actual equipment to obtain better parameters.

The selection of state vector $\boldsymbol{q}$ has been mentioned in Section 3. Section 1 also shows that the LQR control input in this paper is the torque of the hub motor of the robot. Through the calculation of the current, the function relationship between the two is used to convert it into the torque, and it should be applied to the hub motor so as to achieve the requirements of stable balance control.

In the control program, the robot's initial center of gravity position and target position are given. After the WLR is started, the torque of the hub motor is calculated and sent at each control period to track the target position. In this process, a fixed torque is also applied to the leg joint motor to keep the length of the leg in the initial state. The robot is dynamically adjusted by the following control law:

$$u = -K * \hat{q} \tag{13}$$

where $\hat{q}$ is used as an estimator is obtained by Kalman filter. For linear systems, controllability and observability are separable. According to the analysis, the WLR can maintain stable operation against external interference and reach the target position.

It is also noting that, the change of leg length is introduced in the simulation. In general, the leg length of the robot is fixed. When the robot leg length changes, its center of gravity will also change. This can be regarded as an interference, so that the robot needs to determine the equilibrium position in a new state. The flow chart of the whole balance control process is shown in Figure 4.

5 Simulation Results and Analysis

Numerical simulation tests are carried out in the environment of MATLAB/Simulink. In the simulation environment, the change trend of system state is similar to that in the actual environment. Therefore, the simulation test provides a basis for the actual equipment experiment. Specifically, the following three cases of simulation tests are designed.

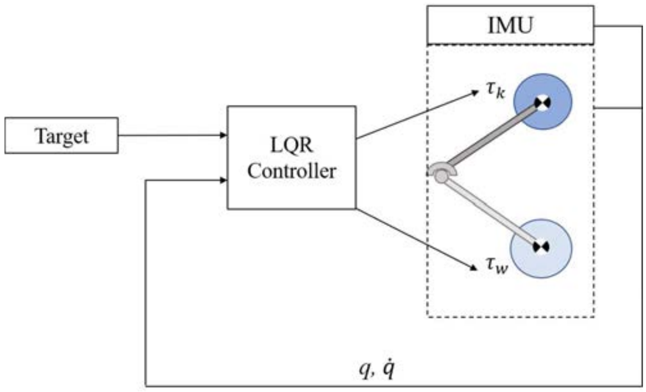

Fig. 4: Flow chart of balance control

Case 1: The simulation test without any interference is carried out. The target position is 10 m away from the initial position. The results are given in Figure 5. At the same time, as the target position is reached, the center of gravity of the robot is restored to the vertical direction.

Case 2: A step disturbance is exerted on the robot after it reaches the target position. The robot moves on an oblique plane of 0.15 radians at this situation. According to Figure 6, it can still recover to the target position quickly, and the adjustment time is about 2.3 seconds.

Case 3: Besides the oblique plane disturbance, the robot's initial inclination angle θ_p is intentionally set as $5°$ to test the robustness of the proposed method. The result is given by Figure 7. It can be seen that after changing the center of gravity position, the time for the robot to reach the target position is 5 seconds, which is basically the same as the time required without changing the center of gravity position.

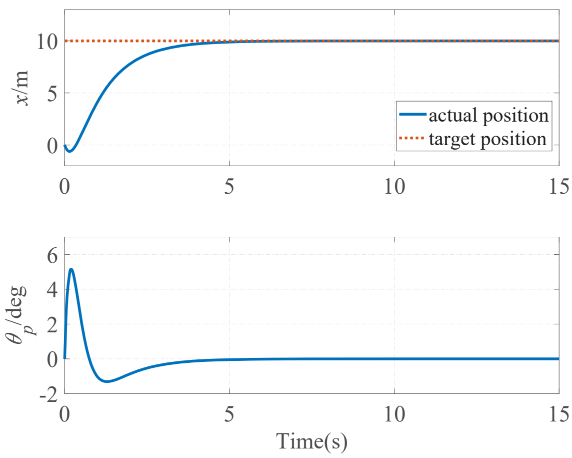

Fig. 5: Simulation results of case 1

6 Actual Experimental Results and Analysis

Based on the previous analysis, the robot model and control strategy are established, and the simulation results of balance control are illustrated. In this section, some experiments are carried out on the actual WLR to test the performance of the control strategy. Table 3 gives the robot param-

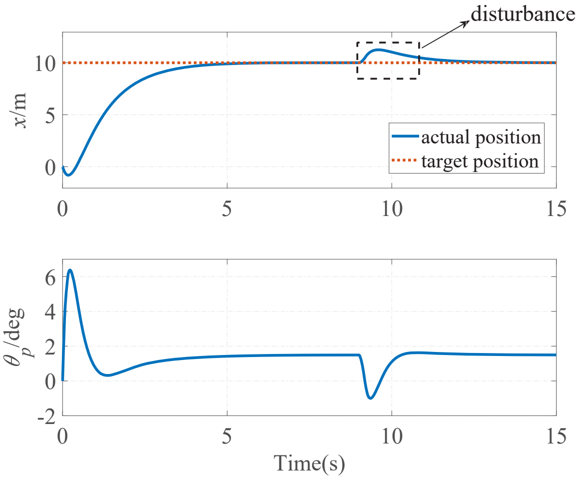

Fig. 6: Simulation results of case 2

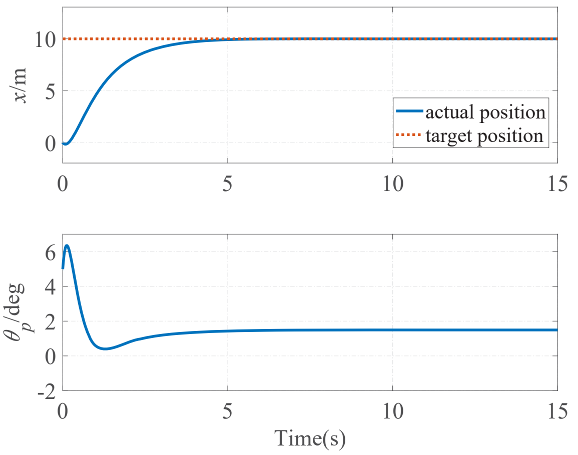

Fig. 7: Simulation results of case 3

eters such as rated velocity, angular velocity, torque, height and so on as known parameters.

Table 3: Rated Parameters of the Robot

Quality	10.25 Kg
Rated Hub Motor Speed	0.5 m/s
Rated Joint Motor Torque	13 Nm
Height of Robot	0.7 m

In actual experiments, two cases are designed to test the robot's tracking speed, accuracy and resistance to interference. The tracking task will be executed only when the robot is ready for moving, which prevents the occurrence of possible accidents.

Case 1: The expected position is 40 radians away from the initial positionthe WLR is not subjected to other disturbances during this process. The results show that the robot takes approximately 13 seconds to reach the target position. After the target position is reached, the torque of the wheel motor remains at an essentially constant value and the centre of gravity of the WLR remains essentially unchanged.

Case 2: After the robot is stabilised in the target position, an unknown disturbance (a backward pull acting on the robot to move it away from the desired position) is added to WLR to observe its resistance to disturbance. From the experimental results, it can be seen that after the disturbances are exerted on the robot, it takes about 8 seconds to return to its original position. Figure 9 also shows that the robot only requires 3 seconds to recover from disturbances.

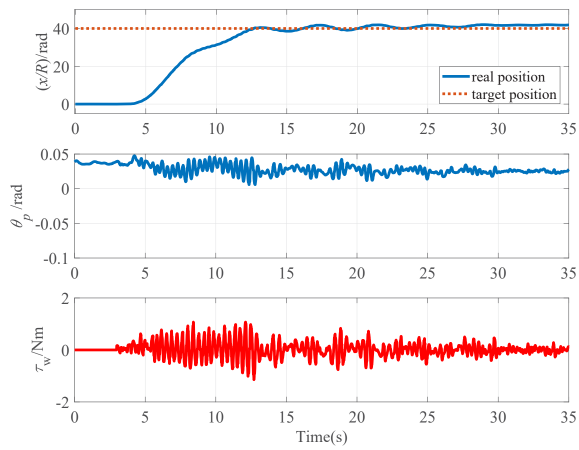

Fig. 8: Experimental results of case 1

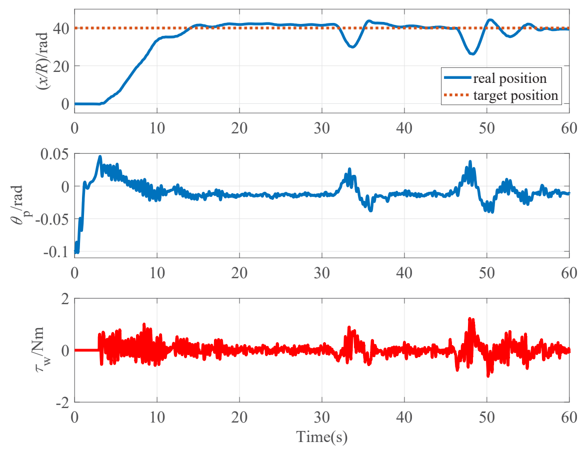

Fig. 9: Experimental results of case 2

The experimental results show that the robot has good stability and can resist unknown interference rapidly. More advanced control strategies will be investigated to solve this problem in our future work.

7 Conclusion

In this paper, a new type of WLR actual platform is introduced. After linearization, its dynamic model can be simplified as an inverted pendulum. Therefore, LQR control strategy is used to realize the balance control of the robot by combining the information obtained by IMU and other sensor data. Through numerical simulation and practical experiments, the results show that it can execute the balance task and has a satisfactory anti-interference ability. The control strategy will be further optimized. Motion planning of the robot will also be investigated to achieve better balance control in more complex scenarios.

Acknowledgement

The authors acknowledge the financial support from Guangdong Provincial Key Lab of Robotics and Intelligent System, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, 518055.

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